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Subject Name : Data Structures and Algorithms with Python
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Credit : 4
Grading Pattern : 1

Lab 5 Assignment Questions

01. Write a menu driven Program in Python to implement both Static and Dynamic Stack for both integer numbers and characters and its operations shown as below:

- 1.Push
- 2.Pop
- 3.Display the top element
- 4.Display all stack elements
- 5.Quit

Integer Number:

An integer is a number with **no decimal or fractional part** and it **includes negative and positive numbers, including zero**. A few examples of integers are: -5, 0, 1, 5, 8, 97, and 3,043.

Static Stack:

Static Stack means number of items in stack or **size of Stack is fixed**. Static stacks **allocate memory at compile time**. You give the size of the Stack at the time of **writing program** in Static Stack.

Dynamic Stack:

Dynamic Stack means number of items in stack or **size of Stack is not fixed**. Dynamic stacks **allocate memory during runtime**. You give the size of the Stack at the time of **running program** in Dynamic Stack.

Enter your choice : 1

Enter the item to be pushed : 1

- 1.Push
- 2.Pop
- 3.Display the top element
- 4.Display all stack elements
- 5.Quit

Enter your choice : 1

Enter the item to be pushed : 2

- 1.Push
- 2.Pop
- 3.Display the top element
- 4.Display all stack elements
- 5.Quit

Enter your choice : 1

Enter the item to be pushed : 3

- 1.Push
- 2.Pop
- 3.Display the top element
- 4.Display all stack elements
- 5.Quit

Enter your choice : 1

Enter the item to be pushed : 4

- 1.Push
- 2.Pop
- 3.Display the top element
- 4.Display all stack elements
- 5.Quit

Enter your choice : 4

Stack elements :

4
3
2
1

- 1.Push
- 2.Pop
- 3.Display the top element
- 4.Display all stack elements
- 5.Quit

Enter your choice : 3

Item at the top is : 4

- 1.Push
- 2.Pop
- 3.Display the top element
- 4.Display all stack elements
- 5.Quit

Enter your choice : 2

Popped item is : 4

- 1.Push
- 2.Pop
- 3.Display the top element
- 4.Display all stack elements
- 5.Quit

Enter your choice : 2

Popped item is : 3

- 1.Push
- 2.Pop
- 3.Display the top element

4.Display all stack elements

5.Quit

Enter your choice : 2

Popped item is : 2

1.Push

2.Pop

3.Display the top element

4.Display all stack elements

5.Quit

Enter your choice : 2

Popped item is : 1

1.Push

2.Pop

3.Display the top element

4.Display all stack elements

5.Quit

Enter your choice : 4

Stack is empty

1.Push

2.Pop

3.Display the top element

4.Display all stack elements

5.Quit

Enter your choice : 3

Stack Underflow

02. Write a Menu Driven Program in Python to implement both Static and Dynamic Queue for Integer numbers and characters and its operations shown as below:

1.Insert

2.Delete

3.Display element at the front

4.Display all elements of the queue

5.Quit

Integer Number:

An integer is a number with **no decimal or fractional part** and it **includes negative and positive numbers, including zero**. A few examples of integers are: -5, 0, 1, 5, 8, 97, and 3,043.

Static Queue:

Static Queue means number of items in Queue or **size of Queue is fixed**. Static queue has a **fixed size allocated at compile time**. You give the size of the Queue at the time of **writing program** in Static Queue.

Dynamic Queue:

Dynamic Queue means number of items in Queue or **size of Queue is not fixed**. Dynamic queue can **grow or shrink in size as needed during runtime**. You give the size of the Queue at the time of **running program** in Dynamic Queue.

Enter your choice : 1

Input the element for adding in queue : 1

1.Insert

2.Delete

3.Display element at the front

4.Display all elements of the queue

5.Quit

Enter your choice : 1

Input the element for adding in queue : 2

1.Insert

2.Delete

3.Display element at the front

4.Display all elements of the queue

5.Quit

Enter your choice : 1

Input the element for adding in queue : 3

1.Insert

2.Delete

3.Display element at the front

4.Display all elements of the queue

5.Quit

Enter your choice : 1

Input the element for adding in queue : 4

1.Insert

2.Delete

3.Display element at the front

4.Display all elements of the queue

5.Quit

Enter your choice : 4

Queue is :

1 2 3 4

1.Insert

2.Delete

3.Display element at the front

4.Display all elements of the queue

5.Quit

Enter your choice : 3

Element at the front is 1

1.Insert

2.Delete

3.Display element at the front

4.Display all elements of the queue

5.Quit

Enter your choice : 2

Deleted element is 1

- 1.Insert
- 2.Delete
- 3.Display element at the front
- 4.Display all elements of the queue
- 5.Quit

Enter your choice : 2

Deleted element is 2

- 1.Insert
- 2.Delete
- 3.Display element at the front
- 4.Display all elements of the queue
- 5.Quit

Enter your choice : 2

Deleted element is 3

- 1.Insert
- 2.Delete
- 3.Display element at the front
- 4.Display all elements of the queue
- 5.Quit

Enter your choice : 2

Deleted element is 4

- 1.Insert
- 2.Delete
- 3.Display element at the front
- 4.Display all elements of the queue
- 5.Quit

Enter your choice : 2

Queue Underflow